Information-limited fidelity estimation of composed quantum channels: when to compose, measure, or learn

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Abstract. Scientific machine-learning estimators should be assessed against the information available at inference and the cost of the reference calculation. We study end-to-end entanglement fidelity for composed quantum channels. Complete Markovian Choi descriptions favour deterministic superoperator composition, which is numerically exact at the dimensions considered here. Our main collision benchmark instead supplies reset-bath marginal channels and omits bath retention. Counterfactual replay of 2048 fixed inputs at 15 retention values gives a mean target range of 0.229. Half of this range gives a mean worst-case error lower bound of 0.115 for every estimator using the same input. The Bayes mean absolute error (MAE) under a uniform high-memory grid is 0.060. An independent exchange-coupled model with noncommuting interaction and bath terms gives a mean range of 0.100 and a corresponding lower bound of 0.050. We then combine a calibrated marginal prediction with a same-query Direct Fidelity Estimation (DFE) measurement. On a fresh 4096-sample split, a 32-shot hybrid reaches MAE 0.0657, close to the 0.0641 of 64-shot DFE. A 64-shot hybrid reaches 0.0539, close to the 0.0525 of 96-shot DFE. Under a 5% symmetric outcome-flip stress test, calibrated hybrids remain below the error of same-budget DFE. The finite-shot calibration costs are recovered after 192 and 256 deployment queries in the two matched-accuracy comparisons. These results support an estimator-selection procedure that begins with a representation audit, then accounts for measurement access and query volume.

Keywords: information-limited learning, scientific machine learning, quantum channels, fidelity estimation, identifiability, non-Markovian dynamics

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1. Introduction

The variables available at inference determine which scientific estimator is appropriate. A deterministic calculation is suitable when those variables determine the target. Partial observations leave uncertainty that requires an additional measurement or a distributional assumption. Composed quantum channels make this issue explicit because per-step descriptions, classical simulators, and measurements of the deployed process provide different information.

We study the end-to-end entanglement fidelity F_{e} of a channel sequence. This quantity appears in repeater and purification design [1, 2], NISQ circuit validation [3], and fidelity-based screening for quantum error correction [4]. Each estimator uses a different source of information. Products of marginal fidelities and Kraus-path Monte Carlo use per-step channel descriptions. Exact superoperator composition applies when these descriptions specify a Markovian process. Direct Fidelity Estimation (DFE) [5] measures the deployed composed channel. A learned sequence model can amortise computation or prior characterisation. Its accuracy remains subject to the information contained in its inputs.

We examine this information constraint with a non-Markovian collision process. The inputs are reset-bath marginal Choi matrices. The target comes from joint propagation of the system and bath, and the input omits the bath-retention variable η. Counterfactual replay keeps every supplied input fixed and varies η. The resulting target values define a set of physically valid answers for the same observed input. Their spread gives estimator-independent lower bounds that can be compared with trained predictors. We repeat the audit with an exchange-coupled model whose interaction and bath terms do not commute. This checks whether the ambiguity is specific to the original dephasing construction.

The same analysis motivates a measurement-conditioned estimator. It forms a constrained convex combination of the marginal prediction and a low-shot DFE estimate from the same query. The fitted weight favours the prior when the pilot has high variance and shifts toward the measurement as the shot budget grows. Calibration labels, calibration pilots, and test measurements use independent binomial outcomes. The cost analysis includes every calibration shot. We generate a fresh OOD dataset with separate random streams for the collision parameters and η. We also test symmetric readout errors and a two-stage allocation rule. These controls probe random-number coupling, measurement noise, and sensitivity to nonuniform shot allocation.

The benchmark includes strong deterministic and sampling controls. Exact composition gives the best result in the hardware-like Markovian regime. Classical Monte Carlo gives the best result in the two-qubit regime when a large sampling budget is available. In the collision regime, DFE enumerates all four Pauli settings and avoids the extra variance from sampling settings with replacement. The learned estimators are assessed within the regimes left open by these controls.

Our contributions are:

(a) We formalise a representation-dependent ambiguity diameter for composed-channel fidelity and derive worst-case and distributional MAE lower bounds that apply to every estimator using the same inputs.

(b) A paired counterfactual collision audit covers 2048 fixed marginal inputs and 15 high-memory values. It measures a mean ambiguity diameter of 0.229 and a Bayes MAE floor of 0.060.

(c) A second audit uses 1024 exchange-coupled sequences with noncommuting interaction and bath terms. It recovers the same information failure with a mean ambiguity diameter of 0.100.

(d) We combine a calibrated marginal prediction with same-query DFE measurements. The cost analysis includes every finite-shot calibration label and pilot. The hybrid approximately halves the per-query shots needed to reach MAE $\simeq 0.065$ and reduces them by one third near MAE $\simeq 0.053$.

(e) The benchmark includes exact composition, low-capacity regressors, Monte Carlo, stratified DFE, independent random streams, symmetric readout errors, and full amortised costs. These controls define when to compose, measure, learn, or combine estimators.

The study focuses on diagnosing information insufficiency and reducing its effect with a costed measurement. Architectural comparisons provide supporting evidence.

2. Setup

2.1. Channel sequences and entanglement fidelity

Let $\mathcal{H}_S$ be a finite-dimensional system Hilbert space with dimension d. A quantum channel is a completely positive trace-preserving (CPTP) map $\Lambda : \mathcal{B}(\mathcal{H}_S) \to \mathcal{B}(\mathcal{H}_S)$. For an ordered sequence $\Lambda_1, \ldots, \Lambda_n$, we write $\Lambda_{1:n} \equiv \Lambda_n \circ \cdots \circ \Lambda_1$. We use the unnormalised Choi-Jamiolkowski matrix [6, 7]

$$\mathcal{J}(\Lambda) \;=\; \sum_{i,j=0}^{d-1} |i\rangle\langle j| \otimes \Lambda(|i\rangle\langle j|) \in \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_S), \tag{1}$$

which obeys $\mathcal{J}(\Lambda) \succeq 0$, $\mathrm{Tr}[\mathcal{J}(\Lambda)] = d$, and $\mathrm{Tr}_2[\mathcal{J}(\Lambda)] = \mathbb{1}_S$ for every CPTP map.

The prediction target is the entanglement fidelity [8] of a channel with respect to the identity,

$$F_{\mathrm{e}}(\Lambda) \;=\; \frac{1}{d^2}\mathrm{Tr}[\mathcal{J}(\mathrm{id})^\dagger \mathcal{J}(\Lambda)] \;=\; \langle \Phi^+|(\mathrm{id} \otimes \Lambda)(|\Phi^+\rangle\langle\Phi^+|)|\Phi^+\rangle, \tag{2}$$

where $|\Phi^+\rangle = d^{-1/2}\sum_i |ii\rangle$. Our downstream estimators and DFE target F_{e}. Average gate fidelity can be recovered from $F_{\mathrm{avg}} = (dF_{\mathrm{e}} + 1)/(d + 1)$.

The composed-channel prediction problem is

$$\textsc{FidelityFormer}_\theta : \quad (\mathcal{J}(\Lambda_1), \ldots, \mathcal{J}(\Lambda_n)) \;\longmapsto\; p_\theta(F_{\mathrm{e}}(\Lambda_{1:n})), \tag{3}$$

where p_θ is a calibrated distribution over $F_{\mathrm{e}} \in [0, 1]$, implemented with a quantile head.

2.2. Representation ambiguity and estimator-independent limits

The central issue can be stated without reference to a particular model. Two physical instances may produce exactly the same estimator input while differing in a hidden variable. If their true fidelities differ, an estimator that sees only the shared input must return the same prediction for both. The separation between the two targets therefore creates an error floor before any model is trained.

We formalise this observation as follows. Let z denote a complete physical instance, $F(z) \in [0,1]$ its target fidelity, and $x = \phi(z)$ the representation supplied to an estimator. The indistinguishable set $\mathcal{A}_\phi(x) = \{z : \phi(z) = x\}$, also called a fibre, contains all physical instances that produce the same input x. Define the range of possible targets within this set as

$$\Delta_\phi(x) = \sup_{z,z' \in \mathcal{A}_\phi(x)} |F(z) - F(z')|. \tag{4}$$

We call $\Delta_\phi(x)$ the representation ambiguity diameter. For any deterministic estimator $h(x)$,

$$\sup_{z \in \mathcal{A}_\phi(x)} |h(x) - F(z)| \geq \frac{1}{2}\Delta_\phi(x). \tag{5}$$

To see why, choose two instances whose targets lie near the two ends of the range. The prediction $h(x)$ is shared by both instances. The triangle inequality then gives $\Delta_\phi(x) \leq |F(z) - h(x)| + |h(x) - F(z')|$, so at least one error must be half the range or larger. This is a worst-case statement for each observed input. To obtain an average MAE floor, we must also specify how likely the possible targets are. Under a conditional distribution $p(F \mid x)$, the conditional median minimises MAE and gives the Bayes risk

$$R^\star_\phi = \mathbb{E}_x \mathbb{E}_{F|x}[|F - \mathrm{median}(F \mid x)|]. \tag{6}$$

We estimate the two floors by replaying fixed collision parameters across a grid of hidden retention values. The first floor describes the worst case within each indistinguishable set. The second describes average error under the chosen distribution over retention values. The supplementary material gives the protocol and proof details.

2.3. Three operating regimes

We evaluate the estimators in three regimes spanning more than two orders of magnitude in per-channel infidelity.

(R1) Device regime (d=2, $\varepsilon \approx 10^{-3}$). Single-qubit channels are sampled from parametric noise models spanning amplitude damping, phase damping, depolarising, and Lindblad families. The numerical ranges are centred near a per-channel infidelity of $\varepsilon \approx 10^{-3}$ and define a synthetic low-noise benchmark. Qiskit provides the channel representations and simulation tools used in this construction [9]. The `family_ood` split holds out the Pauli-channel family entirely.

(R2) Non-Markovian collision regime (d=2, $\varepsilon \approx 0.4$). The system and a bath qubit initially in $|+\rangle$ interact through step-dependent commuting ZZ Hamiltonians. After each collision, the joint state is retained with probability η. The bath is reset in the remaining cases. Training, calibration, and ID data use $\eta \in [0, 0.7]$. The legacy `family_ood` file contains the stronger interval $\eta \in [0.85, 0.99]$. This split changes memory strength while keeping the collision family fixed.

An independent representation audit replaces this interaction with excitation exchange and uses a diagonal bath state. Its exchange and local bath-precession terms do not commute. This family is used to test the ambiguity claim and is not used to train or select the neural model.

(R3) Two-qubit order-sensitive regime (d=4). This regime contains two-qubit channels built from correlated dephasing, two-qubit depolarising noise, and imperfect SWAP/CNOT operations with both coherent and incoherent errors. The order-sensitive variant introduces non-commutativity between consecutive channels, so order-invariant baselines such as DeepSets lose relevant information. This regime tests whether the behaviour observed at $d = 2$ extends to larger systems.

The three regimes answer different questions. R1 is a negative control in which complete Markovian inputs should favour direct composition over learning. R2 is the central information test and asks what can be estimated when the input omits bath memory. R3 tests approximate estimators under richer, order-sensitive two-qubit dynamics.

2.4. Splits and metrics

Each regime contains `train`, `calib`, `id_test`, `length_ood`, and distribution-shift splits. The first three use lengths in $\{2, 4, 8, 16\}$. The length split uses $\{24, 32, 48\}$. R1 and R3 hold out a noise family. R2 holds out a high-η interval and keeps the legacy filename `family_ood` for code compatibility.

The main metric is pooled MAE on F_e across five independent seed-controlled splits. We also report MAE after post-hoc affine recalibration, as described in Appendix Appendix B, and count the quantum shots used by measurement-based estimators. The supplementary material reports CRPS, expected calibration error (ECE), and quantile coverage. Tables give means over five seeds unless specified otherwise. Standard deviations accompany the neural results, and the appendix lists the recalibration coefficients for each seed.

3. Architecture

The model below supplies the marginal prediction used both as a zero-shot estimator and as the prior in the hybrid. The representation audit and measurement procedure apply to other predictor architectures as well.

3.1. Choi-aware per-channel encoder

Each channel Λ_i enters the model through its Choi matrix $\mathcal{J}(\Lambda_i) \in \mathbb{C}^{d^2\times d^2}$. This representation preserves positivity and the partial-trace constraint. We Hermitian-symmetrise $\mathcal{J}(\Lambda_i) \to \frac{1}{2}(\mathcal{J}(\Lambda_i) + \mathcal{J}(\Lambda_i)^\dagger)$ and separate the real and imaginary parts. The resulting feature has $2d^4$ real values per channel. It contains 32 values for d=2 and

512 for d=4. The imaginary block is anti-symmetric and includes redundant entries. We keep this layout for a simple implementation. A two-layer multilayer perceptron with width $D = 256$ and GELU activations maps the feature to a channel token $z_i \in \mathbb{R}^D$. The PTM encoder in Supplementary Sec. S4 uses 16 real values per single-qubit channel. Its pooled MAE differs from the Choi encoder by less than 0.003 on every collision split.

3.2. Causal transformer backbone

A depth-6 causal transformer processes the channel tokens $z_1, \dots, z_n$. It uses eight attention heads, model width 256, a fourfold feed-forward expansion, and rotary positional encoding. Causal masking prevents the token at position i from attending to later channels. This choice gives more stable length-OOD behaviour. Bidirectional attention performs slightly better on `id_test` and gives similar results after OOD calibration, as reported in Section 5. We use causal masking as a stable default that follows the order of channel composition.

3.3. Quantile distribution head

The pooled sequence representation feeds two heads. The primary head produces $K = 9$ quantile predictions at nominal levels $\{0.1, 0.2, \dots, 0.9\}$, trained with the pinball (quantile-regression) loss

$$\mathcal{L}_{\text{pinball}} \; = \; \sum_{k=1}^{K} \rho_{q_k}(F_{\text{e}}^{\text{true}} - \widehat{F}_{q_k}), \tag{7}$$

with $\rho_q(u) = u(q - \mathbb{1}[u < 0])$. We initialise the median-quantile bias to give a softplus output near $F_{\text{e}} = 0.7$. Lower initial values led to saturation and small gradients during the first few hundred updates on deep circuits. The ablation in the supplementary material replaces the quantile head with a truncated-Gaussian head that outputs μ and $\log \sigma$.

3.4. Auxiliary physics-consistency loss

A small auxiliary head predicts the purity $\text{Tr}[\mathcal{J}^2]/d^2$ of the composed Choi. Its loss weight is $\lambda_{\text{aux}} = 0.1$. The task encourages the representation to distinguish nearly unitary compositions from strongly decoherent ones. Setting λ_{aux} to zero produces MAE within seed variation on all three collision splits. We keep the head for consistency with the reported training runs and remove it at inference. The supplementary material gives the full ablation.

3.5. Post-hoc affine recalibration

Distribution shift between training and OOD splits manifests as a near-affine bias in the predicted median: $\widehat{F}^{\text{raw}}_{q_{0.5}} \approx a^{-1}(F_{\text{e}}^{\text{true}} - b)$ over the OOD population. We exploit this with a one-line correction

$$\widehat{F}^{\mathrm{cal}}_{q_{0.5}} \;=\; a \cdot \widehat{F}^{\mathrm{raw}}_{q_{0.5}} + b, \tag{8}$$

Ordinary least squares fits a and b on a labelled OOD calibration set that is disjoint from the test set. We use $N_{\mathrm{cal}} = 64$, where the calibration-size sweep begins to saturate. The supplementary material reports sizes down to eight. Test instances have no labels at inference. The labelled calibration set is collected once before deployment. Synthetic labels come from exact Choi composition in the Markovian regimes and joint propagation of the system and bath in the collision regime. A hardware deployment would obtain these labels from DFE, tomography, or another characterisation procedure. The corresponding cost is paid offline and amortised over later queries.

We choose a linear correction because the residual after fitting a and b is small relative to the OOD bias, as shown in Appendix Appendix B. The same correction is applied to every predicted quantile.

Across five seeds, the fitted slopes have a standard deviation near 0.09. The post-calibration MAE has a standard deviation of 1.3×10^{-3}. Slope and intercept co-vary, so different affine coefficients produce similar calibrated errors. The detailed coefficients appear in Appendix Appendix B.

4. Baselines

We compare FidelityFormer with analytic estimators, measurement-based estimators, classical Monte-Carlo sampling, and neural alternatives.

4.1. Analytic baselines

Product approximation. The primary analytic baseline is

$$\widehat{F}_{\mathrm{prod}} = \prod_i F_{\mathrm{e}}(\Lambda_i), \tag{9}$$

computed directly from the per-channel Choi matrices. It uses only marginal channel information and requires no quantum measurements.

Exact composition of observed channels. When full per-step Choi matrices are supplied and the process is Markovian, their superoperators can be multiplied deterministically and the target fidelity evaluated exactly. We include this baseline and its wall-clock cost. In R2 it composes only the reset-bath marginals, so its residual error measures the information missing from those inputs.

Fuchs–van de Graaf bound. We also use a conservative estimator derived from the relation between fidelity and Choi trace distance,

$$F_{\mathrm{e}} \geq 1 - \frac{1}{2}\|\mathrm{id} - \Lambda\|_{1,1}.$$

Per-channel infidelities are propagated through a sub-additive composition rule to obtain a one-sided lower bound on the composed-channel fidelity. We report its MAE as a point predictor to quantify how well this cheap analytic lower bound tracks the true F_{e}.

Diamond-norm SDP. As a stronger analytic baseline, we solve the Watrous diamond-norm SDP [10] for the marginal-composed channel and convert the result using

$$F_{\mathrm{e}} \geq 1 - \frac{1}{2}\|\mathrm{id} - \Lambda\|_{\diamond}.$$

The SDP is solved with SCS through `cvxpy` [11] and the QuTiP wrapper [12]. Like the product approximation, it operates on the channel obtained by composing the observed marginals. The retained bath remains outside its input.

4.2. Measurement and sampling baselines

Direct Fidelity Estimation [5]. DFE estimates $F_{\mathrm{e}}(\Lambda_{1:n})$ from Pauli measurements on the deployed composed channel. All four Pauli settings have nonzero relevance for the single-qubit identity target. Sampling settings with replacement adds protocol variance once the number of draws exceeds four. We enumerate the four settings and allocate a fixed total shot budget in proportion to the square root of their target relevance. Measurement outcomes follow exact binomial $\{+1, -1\}$ statistics. R2 uses total budgets from 4 to 2000 shots per sequence.

Measurement-conditioned prior and DFE fusion. A marginal predictor has no observation of the realised bath memory. We add an independent B-shot DFE pilot from the same query and combine it with the calibrated prior $h_\phi(x)$:

$$\widehat{F}_{\mathrm{hyb},B} = \operatorname*{clip}_{[0,1]}\left[(1 - w_B)h_\phi(x) + w_B\widehat{F}_{\mathrm{DFE},B}\right], \qquad 0 \leq w_B \leq 1. \tag{10}$$

Least squares fits w_B on 64 OOD calibration instances. Independent 64-shot DFE estimates provide their target labels. Separate B-shot outcomes provide the pilot values used in the fit. The one-time quantum cost is $64 \times 64 + 64 \times B$ shots. Deployment uses B shots per query. Neural, summary-ridge, and exact-marginal priors follow the same protocol.

Monte-Carlo Kraus sampling. This classical baseline samples K Kraus paths from the per-channel marginal channels, propagates the maximally entangled state along each path, and averages the resulting fidelities [13]. We sweep $K \in \{10, 100, 1000\}$. Since the sampling distribution is built from marginal channels, this estimator is blind to correlations carried by a shared bath.

Randomized benchmarking. Randomized benchmarking [14] estimates average gate fidelity from twirled Clifford sequences. Our target is the entanglement fidelity of a fixed composed channel, and our cost model is defined per query. Section 7 discusses

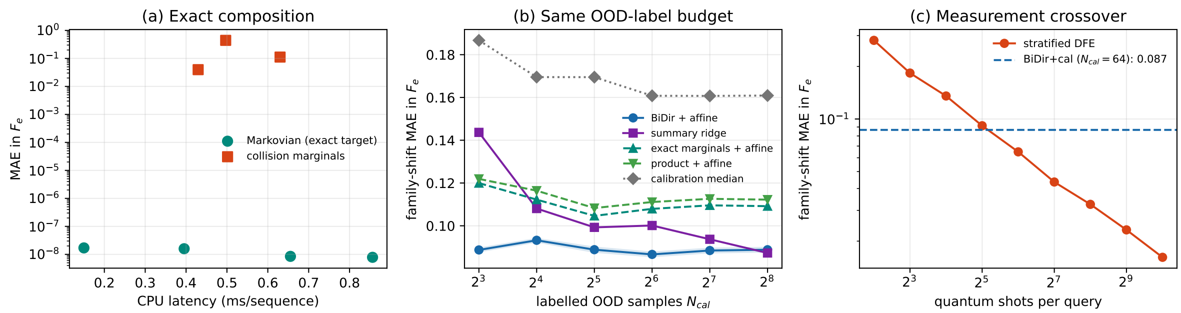

Figure 1. Negative controls and corrected cost-accuracy comparison. (a) Direct composition is numerically exact on Markovian full-Choi inputs at the studied $d \leq 4$ scale. Collision points use observable reset-bath marginals and retain an error from missing memory. (b) All out-of-distribution calibrated methods receive identical labelled indices. (c) Stratified Direct Fidelity Estimation reaches the calibrated neural error near 32 shots per query and gives a lower error at 64 shots.

randomized benchmarking. The numerical cost comparison covers estimators of the fixed-channel target.

4.3. Neural baselines

We train four neural alternatives using the same data, channel encoder, distribution head, optimiser, schedule, and seed list as FidelityFormer.

MLP. A positional MLP that concatenates the Choi features from all channels and preserves their order.

DeepSets [15]. An order-invariant model that sum-pools channel tokens. We include it to test whether the regime contains information that depends on channel order.

BiDir Transformer. A transformer with the same backbone as FidelityFormer and full bidirectional attention isolates the effect of causal masking.

GNN. A linear-chain graph neural network with channel tokens as nodes and edges between consecutive channels. We use two message-passing variants, gnn and generic_gnn, both built on the shared channel encoder.

All neural models are trained with AdamW, learning rate 3×10^{-4}, cosine decay, 80 epochs, batch size 256, and early-stopping patience 20. The neural baselines differ only in sequence backbone and order handling.

5. Experiments

We present the results by operating regime. Figure 1 gives the cost-accuracy summary, and Tables 1, 3 and 4 report the corresponding MAE values.

5.1. Device regime: hardware-realistic noise

Table 1 reports the device-noise results. The input contains the full Choi matrix for every Markovian step, so direct superoperator composition provides the relevant reference. Its MAE is 2.4×10^{-8} on ID, 4.2×10^{-8} on length OOD, and 3.8×10^{-8} on family OOD. These values are consistent with roundoff from float32 features. Direct composition is the appropriate estimator when the full input and a small matrix multiplication are available. The product approximation is the cheapest closed-form alternative and reaches the 10^{-4} to 10^{-3} scale. The near-zero diamond-SDP value on family OOD follows from the Pauli geometry of that split and is specific to this case.

Table 1. Device regime: pooled mean absolute error on F_e. Exact composition uses 1024 sequences per split. Neural rows use five seeds and labelled out-of-distribution calibration. Stored float32 precision sets the error floor for the full-Choi exact baseline.

Method	`id_test`	`length_ood`	`family_ood`
Deterministic and analytic, zero shots			
Exact composition	$\mathbf{2.4\times10^{-8}}$	$\mathbf{4.2\times10^{-8}}$	$\mathbf{3.8\times10^{-8}}$
Product approx.	1.0×10^{-4}	1.6×10^{-3}	2.8×10^{-4}
FvG bound	3.5×10^{-4}	5.4×10^{-3}	1.2×10^{-3}
Diamond SDP	1.0×10^{-2}	5.1×10^{-2}	$7.8\times10^{-8\,\dagger}$
Neural, no per-query quantum measurements after labelled-OOD calibration			
FidelityFormer+ cal	$(4\pm2)\times10^{-4}$	$(4.3\pm0.1)\times10^{-2}$	$(1.1\pm0.2)\times10^{-2}$
MLP + cal	$(6.8\pm0.9)\times10^{-3}$	$(5.0\pm1.4)\times10^{-2}$	$(1.4\pm0.6)\times10^{-2}$
DEEPSETS + cal	$(1.19\pm0.00)\times10^{-2}$	$(6.4\pm0.0)\times10^{-2}$	$(2.2\pm0.0)\times10^{-2}$
BIDIR + cal	$(7\pm2)\times10^{-4}$	$(3.5\pm0.2)\times10^{-2}$	$(1.5\pm0.3)\times10^{-2}$
Measurement and sampling baselines			
MC, K=10 (0 q-shots)	2.6×10^{-2}	6.9×10^{-2}	4.8×10^{-2}
MC, K=100 (0 q-shots)	9.7×10^{-3}	2.1×10^{-2}	1.5×10^{-2}
MC, K=1000 (0 q-shots)	3.1×10^{-3}	6.7×10^{-3}	4.8×10^{-3}

†The held-out set in this split contains a Pauli-channel test case. Pauli channels preserve their structure under composition, and the relevant Fuchs-van de Graaf and diamond-norm relation is saturated up to solver tolerance. The near-zero value follows from this split geometry.

For nearly Pauli-diagonal channels, the leading correction missed by the product approximation scales as $\varepsilon^2 L^2$. With $\varepsilon \approx 10^{-3}$ and $L = 16$, the scale is 2.5×10^{-4}. It is comparable to the observed residual of the product approximation and smaller than the finite-sample fitting error of the neural models. The diagnostic selects the analytic estimator for this regime before model training.

5.2. Non-Markovian collision regime

The collision benchmark isolates an information failure. Its input contains reset-bath marginal Choi matrices. The label comes from joint propagation of the system and bath. For fixed collision parameters, changing the omitted retention η leaves the input tensor unchanged to machine precision and changes the target fidelity.

Paired counterfactual certificate. We draw 2048 base sequences and replay each at 15 equally spaced values of η from 0.85 to 0.99. The supplied marginals remain identical within numerical precision. The mean target diameter is 0.2292, which gives a mean pointwise minimax lower bound of 0.1146. Assigning equal probability to the 15 retention values gives a conditional-median Bayes MAE of 0.0600. The first value concerns the worst case within each indistinguishable input group. The second averages over the specified distribution. We also fit the bidirectional model with one realised retention label for each of 64 calibration inputs. Across 100 combinations of models and calibration draws, its MAE is 0.0902 ± 0.0018. The excess above the Bayes floor is 0.0302. The comparison attributes a substantial part of the residual error to unobserved bath memory.

Table 2. Counterfactual information audits. Both families use 15 retention values in $[0.85, 0.99]$. The model receives the same marginal input at every value within a group.

Quantity	ZZ dephasing	exchange coupling
Fixed marginal inputs	2048	1024
Maximum input change across η	0.0	0.0
Mean diameter $\mathbb{E}[\Delta_\phi]$	0.2292	0.1000
Mean minimax floor $\mathbb{E}[\Delta_\phi/2]$	0.1146	0.0500
Uniform-η Bayes MAE floor $R^\star_\phi$	0.0600	0.0247

Independent noncommuting audit. The exchange model uses an excitation-exchange interaction, a dispersive ZZ term, and local bath precession. The exchange and precession terms do not commute. The bath begins in $0.8|0\rangle\langle 0| + 0.2|1\rangle\langle 1|$. Across 1024 independently drawn parameter sequences, the mean ambiguity diameter is 0.1000. It increases from 0.0617 at length 8 to 0.1372 at length 24. Exact composition agrees with joint propagation at $\eta = 0$ to within 5.6×10^{-16}, while every reset-bath marginal remains unchanged across the retention grid. The omitted-memory limitation therefore persists outside the commuting dephasing model. The supplementary material gives the Hamiltonian and length-resolved results.

Measurement-conditioned recovery. We generate a fresh OOD split of 4096 instances with independent random streams for the physical parameters and η. A set of 64 examples is used to fit the prior and fusion weight. The remaining 4032 examples form the test set. Five neural checkpoints and five measurement repeats give 25 neural-hybrid runs. Independent 64-shot DFE outcomes provide the finite calibration labels. The 32-shot hybrid reaches MAE 0.0657, compared with 0.0641 for 64-shot DFE. The

64-shot hybrid reaches 0.0539, compared with 0.0525 for 96-shot DFE. The summary-ridge hybrid approaches the neural hybrid at high shot counts. Its error is higher at low and intermediate budgets, where the prior receives more weight.

Table 3. Collision out-of-distribution mean absolute error on the independent-random-stream split. Hybrid rows include finite 64-shot calibration labels. Uncertainties are standard deviations over measurement repeats and, for neural rows, five checkpoints.

Method	query shots	offline shots	MAE
BiDir marginal prior	0	4096	0.0885 ± 0.0041
Summary-ridge prior	0	4096	0.0996 ± 0.0041
Stratified DFE	32	0	0.0909 ± 0.0012
Stratified DFE	64	0	0.0641 ± 0.0005
Stratified DFE	96	0	0.0525 ± 0.0004
BiDir hybrid	32	6144	0.0657 ± 0.0019
BiDir hybrid	64	8192	0.0539 ± 0.0012
BiDir hybrid	128	12288	0.0424 ± 0.0016
Summary-ridge hybrid	32	6144	0.0683 ± 0.0027
Summary-ridge hybrid	64	8192	0.0557 ± 0.0024

Nonideal measurements and allocation. We apply an independent symmetric sign flip to each sampled Pauli outcome with probability up to 5%. A calibrated linear inversion corrects the known attenuation before fusion. At 5% readout error, the 32-shot and 64-shot hybrids give MAE 0.0712 and 0.0623. Their clean controls are 0.0650 and 0.0533. Same-budget DFE with the same correction gives 0.1128 and 0.0807. Linear inversion increases variance, so it does not improve finite-shot DFE in every setting. Fusion limits this variance by reducing the fitted measurement weight.

A two-stage allocation experiment gives every query an 8-shot pilot, then assigns either 16 or 48 total shots at an average budget of 32. Ranking queries by a combination of predictive width and prior-pilot disagreement reduces hybrid MAE from 0.0660 to 0.0648. A two-way cluster bootstrap over checkpoints and measurement repeats gives a 95% interval of $[-0.00312, 0.00136]$ for the change. At an average budget of 64, the point estimate is worse than uniform allocation. We treat this as a boundary result rather than evidence for a general adaptive advantage. Full readout and allocation sweeps appear in the supplementary material.

Relation to architecture. The bidirectional transformer gives the strongest marginal prior in the collision experiment. Ridge and exact-marginal priors also improve after fusion. Differences between priors are largest at small measurement budgets, when the fitted weight relies more heavily on the prior. At larger budgets, the DFE pilot dominates and the results converge.

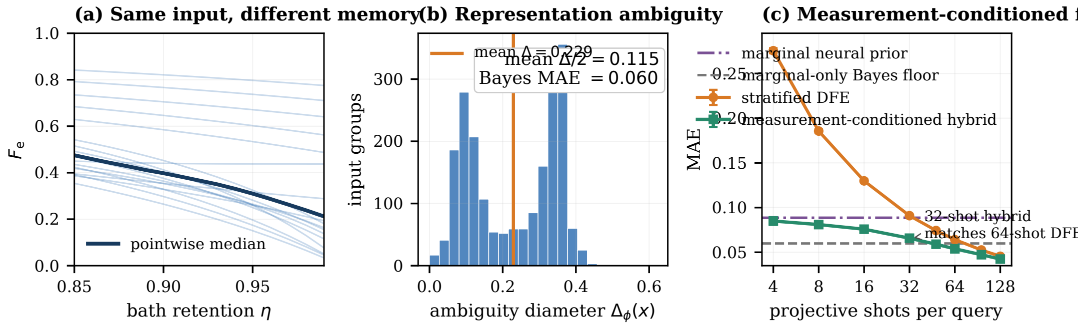

Figure 2. Diagnosing and repairing missing information. (a) Fixed marginal inputs can correspond to different fidelities as hidden bath retention varies. Curves show representative input groups. (b) The empirical ambiguity distribution yields a mean diameter of 0.229, a mean pointwise minimax floor of 0.115, and a uniform-η Bayes mean absolute error floor of 0.060. (c) A same-query Direct Fidelity Estimation pilot supplies information about the deployed process. In the highlighted regimes, the neural hybrid reaches the accuracy of this estimator using roughly one half to two thirds of its per-query shots. Error bars show run-to-run standard deviations.

5.3. Two-qubit $d=4$ scaling

R3 is Markovian and supplies complete per-step Choi matrices. Exact composition therefore remains the reference estimator. It reaches the 10^{-8} numerical floor on every split and takes 0.51–0.83 ms per sequence on the test server. This result fixes the interpretation of R3. The remaining rows compare approximate estimators for cases where exact composition is deliberately unavailable or its input is not exposed to the estimator.

Among these approximate methods, FidelityFormer+cal improves over the product approximation by factors of 1.9, 2.3, and 1.8 on `id_test`, `length_ood`, and `family_ood`. Order-aware neural models also outperform the two closed-form approximations. They do not improve on exact composition.

Classical Monte Carlo gives lower error than the learned models when a sufficient marginal-simulation budget is available. MC with $K = 100$ reaches MAE 0.021 on `id_test`, and $K = 1000$ reduces it to 0.007. These rows map the cost-accuracy tradeoff among approximate methods. They do not replace the deterministic result when full Markovian Choi matrices are available.

Higher-dimensional regime pattern. The $d = 4$ results repeat the main negative control at a larger dimension. Complete Markovian inputs favour direct composition. If only approximate estimators are permitted, richer order-sensitive families increase the error of product and bound-based formulas. Monte Carlo can then outperform the learned models when its sampling budget is affordable.

Table 4. d=4 **two-qubit order-sensitive regime: pooled mean absolute error on F_e.**

Method	`id_test`	`length_ood`	`family_ood`
Exact composition	1.4×10^{-8}	8.3×10^{-9}	1.1×10^{-8}
Product approx.	0.110	0.076	0.091
FvG bound	0.238	0.102	0.218
FidelityFormer + cal	$\mathbf{0.059 \pm 0.002}$	0.033 ± 0.000	0.051 ± 0.001
mlp + cal	0.075 ± 0.001	0.038 ± 0.001	0.060 ± 0.001
deepsets + cal	0.135 ± 0.002	0.038 ± 0.001	0.107 ± 0.002
bidir + cal	0.074 ± 0.000	$\mathbf{0.032 \pm 0.001}$	0.055 ± 0.002
gnn + cal	0.077 ± 0.003	0.033 ± 0.001	0.047 ± 0.001
generic_gnn + cal	0.072 ± 0.003	0.033 ± 0.001	$\mathbf{0.039 \pm 0.002}$
MC, K=10	0.069	0.056	0.070
MC, K=100	0.021	0.021	0.022
MC, K=1000	0.007	0.007	0.007

GNN under family shift. The generic_gnn model gives the lowest `family_ood` MAE in this regime. Its MAE is 0.039, compared with 0.051 for FidelityFormer. This single split suggests that a path-graph inductive bias can help under some family shifts. Broader architectural comparisons require additional shifts.

5.4. Sample-complexity crossover

The cost comparison separates measurements made once during calibration from measurements made for every deployment query. The marginal neural prior uses 4096 offline shots and reaches MAE 0.0885 without query-time measurements. Stratified DFE has no offline measurement cost and uses B shots for each query. Its error decreases approximately as $B^{-1/2}$. The hybrid pays both an offline calibration cost and a smaller per-query measurement cost. Training-set simulation is a separate classical cost in this benchmark.

We compare methods at two similar accuracy levels. First, the 32-shot hybrid reaches MAE 0.0657, while 64-shot DFE reaches 0.0641. The hybrid saves 32 shots per query and has an offline cost of 6144 shots after counting finite-shot labels and independent fusion pilots. It recovers this cost after $6144/32 = 192$ queries. Second, the 64-shot hybrid reaches MAE 0.0539, while 96-shot DFE reaches 0.0525. This hybrid also saves 32 shots per query. Its offline cost of 8192 shots is recovered after $8192/32 = 256$ queries.

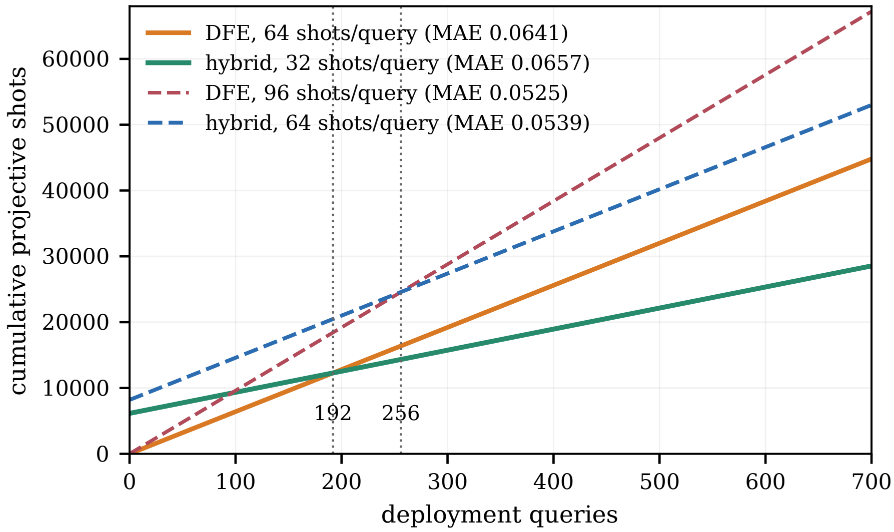

Figure 3. Full amortised measurement cost. The build cost includes 64 independent 64-shot target labels and 64 independent B-shot fusion pilots. The 32-shot hybrid and 64-shot Direct Fidelity Estimation have equal total cost after 192 queries at similar mean absolute error. The corresponding crossing for the 64-shot hybrid and 96-shot Direct Fidelity Estimation occurs after 256 queries.

5.5. Wall-clock latency

Table 5 includes the deterministic baseline. On the same server, direct superoperator composition of full Choi inputs is faster than neural inference for the single-qubit ID setting. The two costs are of the same order for $d = 4$. Exact composition also retains numerical accuracy on the Markovian datasets. The measurements use a CPU for exact composition and an RTX 5090 GPU for neural inference. They provide deployment-scale timings for the present implementation and hardware configuration.

Table 5. Per-sequence wall-clock time in milliseconds for $n \leq 16$. Exact composition uses a CPU. Neural inference uses one RTX 5090.

Method	ms/sequence
Product approximation	0.0013 ± 0.0003
Exact composition, $d = 2$	0.150
FidelityFormer (small), $d = 2$	0.43 ± 0.02
Exact composition, $d = 4$	0.655
MC, K=10	3.45 ± 0.10
MC, K=100	30.2 ± 0.36
MC, K=1000	298.9 ± 4.6

5.6. Distribution calibration: ECE and CRPS

Table 6 reports distributional metrics for the collision regime before the labelled-OOD affine correction in Table 3. The MAE column contains pre-affine point errors and agrees with the raw FidelityFormer result up to rounding. ECE and CRPS evaluate the conformally adjusted quantiles. On the in-distribution split, ECE is 0.0042 and CRPS is 0.018. The correction fitted on ID transfers poorly to the shifted splits. ECE rises to 0.39 on length OOD and 0.50 on family OOD. CRPS follows the same deterioration. Labelled-OOD affine recalibration corrects the mean prediction used in Table 3. The full predictive distribution still requires an OOD-specific calibration method.

Table 6. Collision regime: pre-affine distribution metrics. Results for FidelityFormer pool five seeds and 1024 sequences. Mean absolute error is the raw point error before affine calibration. Expected calibration error uses the predicted median band. Continuous ranked probability score uses all nine quantiles after the in-distribution correction.

Split	MAE	ECE	CRPS
`id_test`	0.025	**0.004**	0.018
`length_ood`	0.138	0.394	0.119
`family_ood`	0.383	0.500	0.359

6. Operating Regimes and Implications

The experiments in Section 5 yield an estimator-selection rule. Its numerical thresholds are calibration points for the regimes studied here and can be adjusted for other channel families.

6.1. The $\varepsilon^2 L^2$ scaling argument

For a Pauli-diagonal channel Λ_i with infidelity $\varepsilon_i = 1 - F_{\mathrm{e}}(\Lambda_i)$, the product approximation expands as

$$1 - \widehat{F}^{\mathrm{prod}}(\Lambda_{1:L}) \;=\; 1 - \prod_{i=1}^{L}(1-\varepsilon_i) \;=\; \sum_{i=1}^{L}\varepsilon_i - \sum_{i<j}\varepsilon_i\varepsilon_j + \mathcal{O}(\varepsilon^3 L^3). \tag{11}$$

For Markovian compositions of Pauli-diagonal channels, the true infidelity shares the first-order term $\sum_i \varepsilon_i$. Differences between the true composed fidelity and the product approximation enter through pairwise and higher-order terms. At the pair level this discrepancy scales as $\sum_{i<j}\varepsilon_i\varepsilon_j$, which is of order $\varepsilon^2 L^2$ when the per-channel infidelities are comparable.

For non-Markovian compositions, the same scaling serves as a warning diagnostic. A shared bath can generate a cross-correlation term for each pair of time steps. These terms are absent from a marginal representation, and their number grows like L^2. We use $\varepsilon^2 L^2$ as an order-of-magnitude flag for testing marginal factorisation. Large ε or non-perturbative bath coupling places the expression outside its quantitative range.

Small ε: hardware-like noise. In the device regime R1, $\varepsilon \approx 10^{-3}$. At $L = 16$, the diagnostic scale is $\varepsilon^2 L^2 \approx 2.5 \times 10^{-4}$. This is comparable to the residual error of the product approximation and smaller than the fitting error expected from a practical training set. The analytic baseline remains more accurate than the neural models in this regime. Direct measurement can supply an unbiased experimental estimate when required.

Large ε: bath-mediated noise. In the collision regime R2, $\varepsilon \approx 0.4$. At $L = 16$, the diagnostic scale exceeds the fidelity range and lies outside the perturbative regime. Marginal factorisation should be tested directly in this case. Bath-mediated correlations dominate the errors of marginal estimators in our data. A learned surrogate can provide repeated estimates after labelled-OOD recalibration. Stratified DFE reaches similar MAE with about 32 shots per query. The calibrated surrogate serves deployments that require zero-shot queries after an offline characterisation stage.

We use $\varepsilon^2 L^2$ as a screening diagnostic. Small values indicate little room to improve on marginal estimators. Large values indicate that correlated alternatives should be included in the benchmark. The expression provides an order-of-magnitude signal outside the perturbative regime.

6.2. Bath-retention sweep

Figure 4 varies the bath-retention parameter η. At $\eta = 0$, the bath is reset after each collision and the marginal description is accurate. MC-1000 and the product approximation have MAE 0.009 and 0.021. Their errors increase with bath retention. At $\eta = 0.85$, product and MC-1000 reach 0.298 and 0.318, while FidelityFormer+cal reaches 0.065. At $\eta = 0.99$, the calibrated model reaches 0.211. This endpoint lies near the edge of the calibration support and shows the remaining sensitivity to the shift.

6.3. Decision rule

Table 7 summarises the resulting rule of thumb. The thresholds are calibrated to our three regimes and to the η sweep, and should be adjusted for other datasets.

The non-trivial cases are the middle and lower rows, where the deployment channel is correlated or non-Markovian. The choice is then a tradeoff among classical simulation cost, per-query quantum measurement cost, and one-time labelled characterisation cost. In our experiments, measurement-only DFE is the clean default when its required shot count is affordable. A marginal-only prior is justified when deployment measurements are prohibited and the shift has been characterised. Between them, the conditioned hybrid reduces the shots needed for a target MAE by using the learned predictor as a

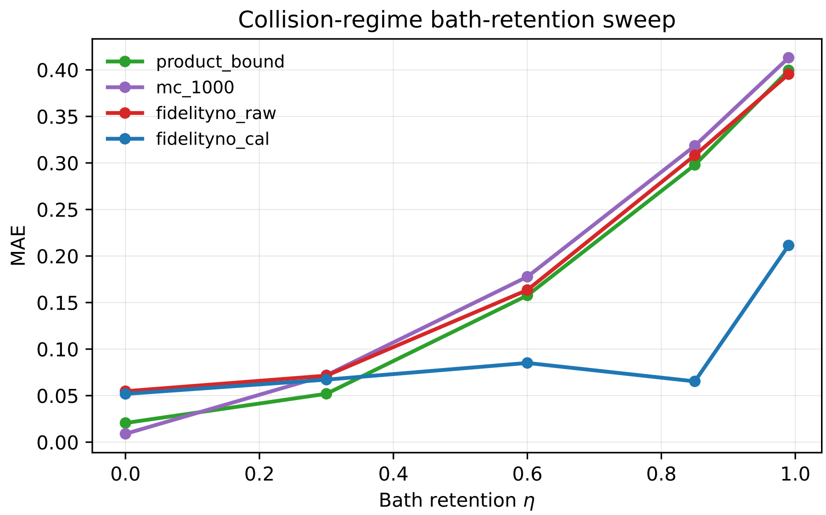

Figure 4. Bath-retention sweep for the decision rule. The retention coefficient η moves from reset-bath dynamics at $\eta = 0$ toward strong memory as η approaches one. Each point uses 1024 sequences. Neural curves average five seeds. Analytic and 1000-sample Monte Carlo curves use the same split.

Table 7. Operating-regime decision rule.

Regime	Diagnostic signal	Recommended estimator
Complete Markovian channel descriptions	full per-step Choi matrices available	exact superoperator composition
Weak correlations / low infidelity	$\varepsilon^2 L^2 \lesssim 10^{-3}$	Product approximation
Deployed channel measurable	query-time quantum shots available	DFE, with shot budget matched to accuracy target
Marginals valid and classical simulation affordable	$\eta \approx 0$ or validated Markovian marginals	MC with $K \in [10^2, 10^3]$
Strong memory, stable shift, and many measurement-free queries	marginals fail and offline labels can be amortised	calibrated low-capacity or neural surrogate
Strong memory without labels or shots	high-fidelity ranking validated on deployment data	raw surrogate for candidate ranking only

shrinkage prior. It becomes cost-effective after the finite calibration expense is amortised over sufficiently many queries.

The uncalibrated BiDir predictions preserve broad ordering on the independent-memory split, with Spearman $\rho = 0.710 \pm 0.002$ and Kendall $\tau = 0.534 \pm 0.002$ across five checkpoints. The predicted highest-fidelity 10% set overlaps the true set by 0.858 ± 0.004. The lowest-fidelity 10% overlap is only 0.144 ± 0.012, close to the random baseline of 0.10. The raw model can therefore support high-fidelity candidate selection in this benchmark. The same evidence does not support low-fidelity failure triage.

6.4. QEC compatibility

Within QEC workflows, FidelityFormer serves as a proxy for repeated queries of composed-channel fidelity. Decoder performance, thresholds, and logical error rates remain separate targets that require decoder-level simulation or experiments.

Logical-channel quality estimation. During code design, a logical-channel fidelity proxy can screen code patches or noise models. The device results favour calibrated analytic estimators for low-infidelity Pauli-like noise. Logical error rates, decoder performance, and threshold behaviour require separate decoder-level simulations or experiments.

Memory cutoff and protocol selection. At higher infidelity, as in the collision regime, the practical question is often whether to keep, abort, or re-purify a noisy memory or partially purified state. In such settings, FidelityFormer+cal can provide repeated fidelity estimates without consuming experimental shots, provided the deployment regime is fixed and labelled calibration data can be obtained offline.

In both cases, the surrogate supplies a preliminary fidelity estimate within a larger QEC workflow. Its practical value depends on whether repeated simulation or measurement costs exceed the cost of the offline labelled set.

6.5. Scope and caveats

Our claims are subject to five limitations.

(i) *Labelled-OOD calibration requires labels.* The headline configuration uses $N_{\mathrm{cal}} = 64$, and the supplementary sweep includes sizes down to eight. These labels come from exact simulation, an offline characterisation campaign, or another labelled source. Their quantum or classical cost is paid once and amortised over later queries. Section Appendix B discusses extensions based on unlabelled distribution matching.

(ii) *The input omits collision memory strength.* For fixed collision parameters, the reset-bath marginals are independent of η. Predictions estimate a correction averaged under the deployment distribution. The realised bath state and retention value remain unobserved. The exchange-coupled audit shows that this issue is not confined to a commuting dephasing Hamiltonian, but both audits use a qubit bath.

(iii) *The $d = 4$ regime uses coherent correlations.* The two-qubit channels in R3 are order-sensitive and use coherent correlations. Bath-mediated systems with $d > 2$ may have different recalibration behaviour and require a separate study.

(iv) *The readout model is controlled but limited.* The robustness audit assumes independent symmetric outcome flips and a known error probability. State-preparation error, asymmetric confusion matrices, drift, and correlated readout faults require a device-specific experiment. The two-stage allocation rule also gives no consistent advantage across the tested budgets.

(v) *The architecture has a secondary role.* FidelityFormer is one member of a broader class of causal sequence models. Several learned backbones reach comparable accuracy within seed variation. The operating-regime analysis and the calibration protocol carry across these architectures.

7. Related work

Fidelity-estimation tools. The estimators in Section 4 are standard tools in quantum information. Product-style approximations appear in repeater-chain and quantum-network analyses [1]. Direct Fidelity Estimation [5] uses Pauli measurements on the implemented channel. Randomised benchmarking [14, 16] estimates Clifford-averaged gate fidelity. Cross-entropy benchmarking [17] provides a related sequence-level diagnostic. Diamond-norm semidefinite programs [10] estimate worst-case channel distance at a higher classical cost.

Learned surrogates for scientific computation. Learned surrogates that replace an expensive numerical estimator with a fast forward model are now widespread across the sciences. Operator-learning architectures such as the Fourier Neural Operator [18] and DeepONet [19] approximate solution maps of parametric differential equations and have become standard tools in scientific machine learning. This literature also studies the conditions under which a surrogate can replace its numerical reference and how its error changes out of distribution. We examine these questions for composed-channel fidelity. The analysis compares learned surrogates with analytic, sampling, and measurement-based estimators and counts the labels used for OOD calibration.

Learned channel-fidelity surrogates. Neural surrogates have been used for quantum-dynamics prediction, dissipative evolution, and Lindbladian learning [20, 21]. Quantum-aware graph networks and equivariant variants have also been explored for quantum-network estimation [22, 23]. In the repeater setting, related work includes reinforcement-learning agents for protocol selection [24] and Bayesian-optimisation approaches to protocol search [25]. Our comparison places these learned surrogates alongside analytic, Monte Carlo, and measurement-based estimators under explicit cost assumptions.

Non-Markovian collision models. Collision models are widely used to study non-Markovian open-system dynamics [26, 27]. Prior work mainly uses them as forward-simulation models. Here we use a collision-model family as a stress test for fidelity estimators that only see marginal channel information.

Distribution shift and post-hoc calibration. Post-hoc calibration methods such as affine, temperature, and isotonic calibration are standard in classification and probabilistic regression [28, 29]. Our labelled-OOD calibration setting is closer to bias correction than to classical coverage calibration: the goal is to remove a systematic shift in the median fidelity prediction. Conformal prediction [30] could be used to calibrate predictive intervals for F_{e} and is a natural extension.

Machine learning for quantum error correction. Neural networks have recently been used to decode syndrome data from near-term surface-code experiments [31]. This task differs from channel-fidelity estimation, but both require a clear account of the information available to the learned estimator and the baseline it replaces. Analytic channel estimators can be sufficient for low-infidelity Pauli-like noise. Learned priors become more useful when correlated noise produces corrections absent from marginal estimators. Decoder-level and threshold-level quantities require separate evaluation.

8. Conclusion

Scientific surrogate modelling begins with the variables available at inference and the cost of existing estimators. For composed-channel fidelity, exact superoperator composition is suitable for complete Markovian descriptions. Stratified DFE provides a direct estimate when a per-query measurement budget is available.

The non-Markovian collision setting is information-limited because the marginal input omits bath retention. Paired counterfactual replay gives a mean fidelity range of 0.229 for identical inputs. Under the uniform high-memory grid, the Bayes MAE floor is 0.060. A separate exchange-coupled audit gives a mean range of 0.100 while keeping the same representation fixed. This second result supports the representation-level interpretation beyond the original dephasing family.

A same-query measurement supplies information about the realised process. Convex fusion of the prior and DFE reaches the accuracy of 64-shot DFE with 32 shots. It also reaches the accuracy of 96-shot DFE with 64 shots on the independent-random-stream split. Full calibration costs amortise after 192 and 256 queries. Under symmetric readout flips, the fitted hybrid reduces its reliance on the noisier measurement and remains more accurate than same-budget DFE at the tested error rates. Nonuniform shot allocation gives mixed results and is not required for the main gain.

Estimator selection follows the available information and query volume. Complete channel descriptions favour composition. Access to the deployed process favours measurement. A learned prior can serve repeated queries under a stable deployment distribution. Combining that prior with a small measurement budget gives intermediate points on the cost-accuracy curve. Counterfactual representation audits make the information assumptions behind these choices explicit.

Author contributions

SW and XL contributed equally to this work. SW and XL conceived the study, developed the methodology, performed the experiments, analysed the results, and prepared the figures and tables. SW drafted the manuscript. WY supervised the project, guided the interpretation and framing of the results, and revised the manuscript. All authors discussed the results, commented on the manuscript, and approved the final version.

Competing interests

All authors declare no financial or non-financial competing interests.

Data availability

All datasets are generated synthetically. The manuscript-specific data generators, evaluation scripts, random seeds, and machine-readable result tables are publicly available at `https://github.com/shchng9963-droid/fidelityno` under the release tag mlst-submission-v2. Training arrays and model checkpoints can be regenerated using the documented commands. An archival DOI will be added when the versioned release is deposited.

Code availability

The source code associated with this manuscript is publicly available at `https://github.com/shchng9963-droid/fidelityno` under the release tag mlst-submission-v2. The tag identifies the exact Git commit, and the software environment is documented in the repository.

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Appendix A. First-order accuracy of the product approximation for Pauli-diagonal channels

We derive the first-order accuracy of the product approximation in (9) for marginals that are Pauli-diagonal in a common basis. This structure describes the IBM `Fake*V2` noise-model regime to leading order. The derivation gives the scaling $\widehat{F}_{\mathrm{err}}^{\mathrm{prod}} = \mathcal{O}(\varepsilon^2 L^2)$.

Let Λ_i act as a Pauli channel,

$$\Lambda_i(\rho) \;=\; \sum_{P\in\{I,X,Y,Z\}^{\otimes \log_2 d}} p_{i,P}\, P\rho P^{\dagger}, \tag{A.1}$$

with $\sum_P p_{i,P} = 1$. The Choi matrix $\mathcal{J}(\Lambda_i)$ is then diagonal in the Pauli basis, and the entanglement fidelity reduces to the identity weight $F_{\mathrm{e}}(\Lambda_i) = p_{i,I}$.

Composition $\Lambda_{1:n} = \Lambda_n \circ \cdots \circ \Lambda_1$ produces another Pauli channel. Its identity weight contains the product $\prod_i p_{i,I}$ and additional terms. Repeated occurrences of the same Pauli can multiply to the identity, as can triples satisfying $P_1 P_2 P_3 = I$. Counting pair contributions (P, P) with $P \neq I$ gives the correction

$$p_{1:n,I} - \prod_{i=1}^{n} p_{i,I} = \sum_{i<j} \sum_{P \neq I} p_{i,P} p_{j,P} \prod_{k \neq i,j} p_{k,I} + \mathcal{O}(L^3 \varepsilon^3). \tag{A.2}$$

Each summand is $\mathcal{O}(\varepsilon^2)$, and there are $\binom{n}{2}$ pairs. For $L = n$ and typical per-channel infidelity ε, the leading correction is of order $\varepsilon^2 L^2$. The product approximation is first-order accurate in ε, with pair terms entering at second order.

Channels outside a common Pauli-diagonal basis introduce an additional Pauli-rotation correction of leading order $\mathcal{O}(\varepsilon^2)$. The collision regime contains non-Pauli dynamics and bath-mediated terms of order $\mathcal{O}(\eta)$. These terms produce the larger correction seen in the collision data.

The experimental claims use measured errors directly. This derivation states the perturbative conditions behind the near-exact product approximation in R1.

Appendix B. Affine recalibration across random seeds

Table B1 reports a legacy audit with separate affine coefficients for each seed. The fit uses a labelled 10% subset of the collision `family_ood` data. This calibration set and split differ from the fixed $N_{\mathrm{cal}} = 64$ audit in Table 2. Its mean MAE of 0.094 forms a separate result from the headline 0.090. Across five seeds, the calibrated MAE has standard deviation 1.3×10^{-3} and the slope has standard deviation about 0.09. Slope and intercept are correlated and are reported together.

Table B1. Affine recalibration coefficients on the collision split. Results for FidelityFormer use five seeds and a labelled 10% out-of-distribution calibration set. The fitted slope is a and the intercept is b in $\widehat{F}^{\mathrm{cal}} = a\widehat{F}^{\mathrm{raw}} + b$.

seed	a	b	MAE raw	MAE cal
0	2.845	−1.916	0.387	0.095
1	2.822	−1.896	0.386	0.095
2	2.918	−1.980	0.387	0.094
3	3.026	−2.080	0.388	0.095
4	3.003	−2.052	0.387	0.092
mean	2.923	−1.985	0.387	0.094
std	0.091	0.081	0.001	**0.001**

Seed-to-seed variation in affine coefficients. The slope varies by about 0.1. Test MAE varies much less because the slope and intercept co-vary over the narrow range of raw predictions. Predictive errors are reported across seeds. The coefficients alone leave the physical source of the OOD bias unresolved.

Quantile calibration. We apply the same affine coefficients to all nine predicted quantiles. This preserves their ordering. On OOD data, empirical coverage can still differ from the nominal level. Medium and high quantiles with $q \geq 0.5$ lie within ± 0.04 of nominal coverage. Low quantiles with $q \leq 0.3$ sometimes over-cover on `family_ood` because predictions are clipped at $F_{\mathrm{e}} = 0$. A per-quantile shift is a possible extension.

Appendix C. Implementation details

Software and numerical backends. Training uses PyTorch 2.6 and CUDA 12.8 on an RTX 5090 with sm_120. Time-dependent Lindblad evolution uses `dynamiqs` 0.3. Choi matrices are cross-validated with `qutip` 5.2 to tolerance 10^{-8}. Diamond-norm SDPs use `cvxpy` 1.9.1 with SCS through `qutip.dnorm`. Hydra manages configurations, and `wandb` records offline logs.

Model architecture. FidelityFormer: $d_{\mathrm{model}} = 256$, depth 6, heads 8, FFN expansion $4\times$, rotary positional encoding, GELU non-linearity, dropout 0.1. Channel encoder: 2-layer MLP, hidden dim 512, GELU. Quantile head: $K = 9$ levels at $\{0.1, 0.2, \ldots, 0.9\}$, softplus output, bias initialised so that the median quantile starts at $F_{\mathrm{e}} \approx 0.7$ (essential to avoid dead-gradient saturation in the deep-circuit regime). Auxiliary physics-consistency head: 2-layer MLP, 0.1 weight on auxiliary loss.

Training protocol. AdamW, lr 3×10^{-4}, cosine decay to 10^{-5}, weight decay 10^{-2}. 80 epochs, batch size 256 (collision and d=4), batch size 128 (device regime). Early-stopping patience 20. We apply a length curriculum: starting epochs sample lengths $\in \{2, 4\}$ uniformly, expanding to $\{2, 4, 8, 16\}$ over the first 30 epochs.

Architecture and training hyperparameters were selected on the device calibration split and then fixed across regimes. This protocol evaluates one shared configuration across the three settings.

Dataset splits. Collision and $d = 4$ splits contain 8000 sequences. Device splits contain 1024 sequences for each backend. Training, calibration, and ID lengths are drawn from $\{2, 4, 8, 16\}$. Length OOD uses $\{24, 32, 48\}$. In the collision data, training uses $\eta \in [0, 0.7]$ and the legacy family_ood file uses $\eta \in [0.85, 0.99]$. Random seeds are disjoint across splits.

Reproducibility and validation. The five seeds are $\{0, 1, 2, 3, 4\}$, applied to (a) NumPy / PyTorch global RNG, (b) channel parameter sampling, (c) train/val/test split assignment, (d) model initialisation, and (e) data shuffling. The 99 unit tests cover CPTP sanity for every channel-family generator, Choi composition associativity, quantile-head monotonicity, recalibration coefficient stability, diamond-SDP correctness against a QuTiP reference, readout-noise sampling, and an end-to-end smoke-train on a 100-sample subset. Total wall-clock for a fresh reproduction on a 2× RTX 5090 setup is approximately 5 hours, dominated by neural training.

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